Method of False Position

The method is also called linear interpolation method or chord method or regula-falsi method. At the start of all iterations of the method, we require the interval in which the root lies. Let the root of the equation $f(x) = 0$, lie in the interval (x_{k-1}, x_k), that is, $f_{k-1}\, f_k < 0$, where $f(x_{k-1}) = f_{k-1}$, and $f(x_k) = f_k$. Then, $P(x_{k-1}, f_{k-1}), Q(x_k, f_k)$ are points on the curve $f(x) = 0$. Draw a straight line joining the points P and Q. We approximate the curve in this interval by the chord, that is, $f(x) \approx ax + b$. The next approximation to the root is given by $x = -b/a$. Since the chord passes through the point *P* and *Q*, we get

$$f_{k-1} = ax_{k-1} + b, \text{ and } f_k = ax_k + b.$$

Subtracting the two equations, we get

$$f_k - f_{k-1} = a(x_k - x_{k-1}), \text{ or } a = \frac{f_k - f_{k-1}}{x_k - x_{k-1}}.$$

The second equation givens $b = f_k - ax_k$.

Hence, the next approximation is given by

$$x_{k+1} = -\frac{b}{a} = -\frac{f_k - ax_k}{a} = x_k - \frac{f_k}{a} = x_k - \left(\frac{x_k - x_{k-1}}{f_k - f_{k-1}}\right) f_k$$

Therefore, in this case, in actual computations, the method behaves like

$$x_{k+1} = \frac{x_k f_1 - x_1 f_k}{f_1 - f_k}, k = 1, 2, \ldots$$

False Position method (regula falsi method) Steps (Rule)

Step 1: Find points x_0 and x_1 such that $x_0 < x_1$ and $f(x_0).f(x_1) < 0$.

Step 2: Take the interval $[x_0, x_1]$ and find next value

$$x_2 = x_0 - f(x_0).\frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

Step 3: If $f(x_2) = 0$ then x_2 is an exact root,

else if $f(x_0).f(x_2) < 0$ then $x_1 = x_2$,

else if $f(x_2).f(x_1) < 0$ then $x_0 = x_2$.

Step 4: Repeat steps 2 & 3 until $f(x_i) = 0$ or $|f(x_i)| \leq$ Accuracy.

Example 1: Locate the intervals which contain the positive real roots of the equation $x^3 - 3x + 1 = 0$. Obtain these roots correct to three decimal places, using the method of false position.

Solution We form the following table of values for the function *f*(*x*).

There is one positive real root in the interval (1, 2).

$$x_0 = 1, x_1 = 2, f_0 = f(x_0) = f(1) = -1, f_1 = f(x_1) = f(2) = 3.$$

$$x_2 = \frac{x_0 f_1 - x_1 f_0}{f_1 - f_0} = \frac{3 - 2(-1)}{3 - (-1)} = 1.25, f(x_2) = f(1.25) = -0.796875.$$

Since $f(1.25)f(2) < 0$, the root lies in the interval (1.25,2).

$$x_3 = \frac{x_2 f_1 - x_1 f_2}{f_1 - f_2} = \frac{1.25(3) - 2(-0.796875)}{3 - (-0.796875)} = 1.407407,$$

$$f(x_3) = f(1.407407) = -0.434437.$$

Since $f(1.407407)f(2) < 0$, the root lies in the interval (1.407407,2).

$$x_4 = \frac{x_3 f_1 - x_1 f_3}{f_1 - f_3} = \frac{1.407407(3) - 2(-0.434437)}{3 - (-0.434437)} = 1.482367,$$

$$f(x_4) = f(1.482367) = -0.189730.$$

Since $f(1.482367)f(2) < 0$, the root lies in the interval (1.482367,2).

$$x_5 = \frac{x_4 f_1 - x_1 f_4}{f_1 - f_4} = \frac{1.482367(3) - 2(-0.18973)}{3 - (-0.18973)} = 1.513156,$$

$$f(x_5) = f(1.513156) = -0.074884.$$

Since $f(1.513156)f(2) < 0$, the root lies in the interval (1.513156,2).

$$x_6 = \frac{x_5 f_1 - x_1 f_5}{f_1 - f_5} = \frac{1.513156(3) - 2(-0.074884)}{3 - (-0.74884)} = 1.525012.$$

$$f(x_6) = f(1.525012) = -0.028374.$$

Since $f(1.525012)f(2) < 0$, the root lies in the interval (1.525012,2).

$$x_7 = \frac{x_6 f_1 - x_1 f_6}{f_1 - f_6} = \frac{1.525012(3) - 2(-0.028374)}{3 - (-0.028374)} = 1.529462.$$

$$f(x_7) = f(1.529462) = -0.010584.$$

Since $f(1.529462)f(2) < 0$, the root lies in the interval (1.529462,2).

$$x_8 = \frac{x_7 f_1 - x_1 f_7}{f_1 - f_7} = \frac{1.529462(3) - 2(-0.010586)}{3 - (-0.10586)} = 1.531116.$$

$$f(x_8) = f(1.531116) = -0.003928.$$

Since $f(1.531116)f(2) < 0$, the root lies in the interval (1.531116,2).

$$x_9 = \frac{x_8 f_1 - x_1 f_8}{f_1 - f_8} = \frac{1.531116(3) - 2(-0.003928)}{3 - (-0.003928)} = 1.531729.$$

$$f(x_9) = f(1.531729) = -0.001454.$$

Since $f(1.531729)f(2) < 0$, the root lies in the interval (1.531729,2).

$$x_{10} = \frac{x_9 f_1 - x_1 f_9}{f_1 - f_9} = \frac{1.531729(3) - 2(-0.001454)}{3 - (-0.001454)} = 1.531956.$$

Now, $|x_{10} - x_9| = |1.531956 - 1.531729| \approx 0.000227 < 0.0005$.

The root has been computed correct to three decimal places. The required root can be taken as $x \approx x_{10} = 1.531956$. Note that the left end point $x = 2$ is fixed for all iterations.

Example 2: Find the root correct to two decimal places of the equation $xe^x = cos\, x$, using the method of false position.

Solution: Define $f(x) = cosx - xe^x = 0$. There is no negative root for the equation. We have $f(0) = 1, f(1) = cos1 - e = -2.17798$.

A root of the equation lies in the interval (0,1). Let $x_0 = 0, x_1 = 1$. Using the method of false position, we obtain the following results.

$$x_2 = \frac{x_0 f_1 - x_1 f_0}{f_1 - f_0} = \frac{0 - 1(1)}{-2.17798 - 1} = 0.31467, f(x_2) = f(0.31467) = 0.51986.$$

Since $f(0.31467)f(1) < 0$, the root lies in the interval $(0.31467,1)$.

$$x_3 = \frac{x_2 f_1 - x_1 f_2}{f_1 - f_2} = \frac{0.31467(-2.17798) - 1(0.51986)}{-2.17798 - 0.51986} = 0.44673,$$

$$f(x_3) = f(0.44673) = 0.20354.$$

Since $f(0.44673)f(1) < 0$, the root lies in the interval $(0.44673,1)$.

$$x_4 = \frac{x_3 f_1 - x_1 f_3}{f_1 - f_3} = \frac{0.44673(-2.17798) - 1(0.20354)}{-2.17798 - 0.20354} = 0.49402,$$

$$f(x_4) = f(0.49402) = 0.07079.$$

Since $f(0.49402)f(1) < 0$, the root lies in the interval $(0.49402,1)$.

$$x_5 = \frac{x_4 f_1 - x_1 f_4}{f_1 - f_4} = \frac{0.49402(-2.17798) - 1(0.07079)}{-2.17798 - 0.07079} = 0.50995,$$

$$f(x_5) = f(0.50995) = 0.02360.$$

Since $f(0.50995)f(1) < 0$, the root lies in the interval $(0.50995,1)$.

$$x_6 = \frac{x_5 f_1 - x_1 f_5}{f_1 - f_5} = \frac{0.50995(-2.17798) - 1(0.0236)}{-2.17798 - 0.0236} = 0.51520,$$

$$f(x_6) = f(0.51520) = 0.00776.$$

Since $f(0.51520)f(1) < 0$, the root lies in the interval $(0.51520,1)$.

$$x_7 = \frac{x_6 f_1 - x_1 f_6}{f_1 - f_6} = \frac{0.5152(-2.17798) - 1(0.00776)}{-2.17798 - 0.00776} = 0.51692.$$

Now, $|x_7 - x_6| = |0.51692 - 0.51520| \approx 0.00172 < 0.005$.

The root has been computed correct to two decimal places. The required root can be taken as $x \approx x_7 = 0.51692$. Note that the left end point $x = 1$ is fixed for all iterations.

H.W

Find a root of an equation $f(x) = x^3 - x - 1$ and the interval (1, 2) using false position method.

Algorithm (False Position)

```
f=@(c)(c)^3-3*(c)+1;
a=0;
b=1;
delta=0.0005;
epsilon=0.0005;
max1=10;
ya=feval(f,a);
yb=feval(f,b);
if ya*yb>0
  disp('Note: f(a)*f(b)>0'),break,end
for k=1:max1
   dx=yb*(b-a)/(yb-ya);
   c=b-dx;
   ac=c-a;
   yc=feval(f,c);
   if yc==0,break;
   elseif yb*yc>0
          b=c;
          yb=yc;
       else
          a=c;
          ya=yc;
        end
     dx=min(abs(dx),ac);
   if abs(dx)<delta,break,end
   if abs(yc)<delta,break,end
end
disp('Regula Falsi Method');
c;
err=abs(b-a)/2;
yc=feval(f,c);
disp(c)
```